

Row IV: which line is BFKL, which is DGLAP, which is finite

The clean three-way split, and two places the five-line form does not yet separate

$$\mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2(x'-y')^2}, \quad \mathcal{R} = \frac{(x-y) \cdot r(x'-y') \cdot r}{(x-y)^2(x'-y')^2 r^2}, \quad r = y' - y, \quad L_r = \log(4\pi^2 \mu^2 r^2), \quad \hat{C} = \frac{\mathcal{P}_{gg}}{2N_c} = \xi \bar{\xi} + \frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi}, \quad \xi_0 = \frac{k^+}{V}$$

The answer in one box. With the measure written in the natural order $\int_{\xi_0}^1 \frac{d\xi}{\xi^2}$, the whole row is

$$\underbrace{\delta(1-\xi) \left[\log \frac{V}{\Lambda} + \log \frac{k^+}{V} \right] \left[\frac{1}{\epsilon} - L_r \right] \mathcal{X}}_{\text{BFKL / JIMWLK}} + \underbrace{\frac{\mathcal{P}_{gg}(\xi)}{2N_c} \left[\frac{1}{\epsilon} - L_r \right] \mathcal{X}}_{\text{DGLAP}} + \underbrace{\xi(1-\xi) [\mathcal{X} - d_\perp \mathcal{R}]}_{\text{finite}}$$

all times $\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} e^{-i\frac{k}{\xi} \cdot r}$ and the LO Wilson-line structure. Your L1 is already exactly the BFKL line. Your L2+L3+L4 rearrange into the DGLAP and finite lines above, with residual 0 (sympy) — but as you have them written the +1 is in the wrong sector, and L3 cannot be written as a convolution with $\frac{d^3 N_{LO}}{d^3 k}$ at all.

1. BFKL / JIMWLK — your line 1

Every signature is present and it needs no rearranging:

- **No ξ -integral.** It is the $\delta(1-\xi)$ term, so $\xi = 1$ throughout — and indeed the phase is $e^{-ik \cdot r}$ with no $\frac{1}{\xi}$, i.e. the hard scale is $k_\perp$, not $k_\perp/\xi$.
- **A rapidity logarithm.** $\log \frac{V}{\Lambda} + \log \frac{k^+}{V} = \log \frac{k^+}{\Lambda}$, one power, from the $\frac{1}{p^+}$ pole. This is the small- x logarithm, not a collinear one.
- **LO colour structure.** $\mathcal{X} [U(y') - U(x')] [U(y) - U(x)] \rho \rho$ is the LO cross section untouched — the mark of an evolution term, which must be (kernel) $\otimes$ (LO).

The bracket $\left[\frac{1}{\epsilon} - L_r \right]$ is the kernel, in dimensional regularisation:

$$\frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2) = \frac{1}{\pi} \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2}.$$

BFKL or JIMWLK? As written it is **JIMWLK**: the Wilson lines are kept to all orders. **BFKL is its dilute limit** — expand $U(y) - U(x)$ to lowest non-trivial order in the target field and the kernel above becomes the BFKL kernel. At the level of this single row the two are the same object; the distinction only shows up once you keep the non-linear terms in the evolution equation.

Your $\vee$ split is doing real work

The two logarithms in line 1 are *not* the same kind of object, which is exactly why splitting them is the right move:

- $\log \frac{V}{\Lambda}$ contains the cutoff Λ . It is **absorbed**, by redefining the target average as the one evolved by JIMWLK from Λ up to V . After that step it is gone from the cross section — it has become the $\vee$ -dependence of the LO term.
- $\log \frac{k^+}{V}$ has no Λ . It **stays**, as a finite NLO remainder, and its $\vee$ -dependence must cancel against the $\vee$ -dependence the evolved LO term just acquired. That cancellation is the check that your factorisation scale is consistent.

2. DGLAP — your lines 2 and 3, after two repairs

The four fragmentation signatures are all there: the measure $\frac{d\xi}{\xi^2}$, the hard factor evaluated at $\frac{k}{\xi}$, the splitting function $\mathcal{P}_{gg}(\xi)$, and the Wilson lines seeing the *parent*. So the labelling is right. Two things are not.

(a) The +1 is in the wrong sector

You have L3 carrying $\frac{\mathcal{P}_{gg}}{2} [1 - L_r]$, i.e. the +1 sits with DGLAP. But L4 still carries $-\left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi} \right] \mathcal{X}$, and

$$\underbrace{\hat{C} \cdot (+1)}_{\text{from L3}} - \underbrace{\left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi} \right]}_{\text{from L4}} = \xi(1-\xi),$$

so **two of the three terms of the +1 are cancelled outright by L4**, and the one that survives, $\xi(1-\xi)\mathcal{X}$, pairs with $-d_\perp \xi(1-\xi)\mathcal{R}$ — it belongs to the finite sector. Lines 3 and 4 as written therefore partially cancel each other: the split is not clean, and the $\mathcal{P}_{gg}$ in L3 is not the whole story of L3. Move the +1 out and DGLAP becomes $\frac{\mathcal{P}_{gg}(\xi)}{2N_c} \left[\frac{1}{\epsilon} - L_r \right] \mathcal{X}$, which is L1's bracket exactly.

(b) Line 3 cannot be a convolution with $d^3 N_{LO}$

DGLAP factorisation requires the coefficient multiplying $\frac{d^3 N_{LO}}{d^3 k} \left(\frac{k}{\xi} \right)$ to be **independent of the transverse integration variables**. L2's coefficient is $\frac{1}{\epsilon}$ — a pure number, fine. L3's coefficient is $[1 - \log(4\pi^2 \mu^2 (y' - y)^2)]$, and $(y' - y)$ is integrated *inside* $\frac{d^3 N_{LO}}{d^3 k}$. So L3 as written is not well defined. The $\log(\mu^2 r^2)$ has to be put back inside $\int_{x,x',y,y'}$, next to $\mathcal{X}$, the way your advisor's earlier version had it.

This is not a nitpick — it is the reason the DGLAP sector here is not yet in factorised form. Only the pure pole factorises; the $\log(\mu^2 r^2)$ does not, because in the CGC the “hard scale” of the collinear logarithm is the dipole size, which lives under the transverse integral.

(c) And the fate of that pole is still open

L2 is the $\overline{\text{MS}}$ fragmentation counterterm. It is a definition, not a physical term. As established before: that $\frac{1}{\epsilon}$ is the *same* $\int d^2 z$ as L1's, it is infrared ($|z| \rightarrow \infty$; the $z \rightarrow y$ region converges), and it is ξ -independent — so it cannot be half-absorbed into a fragmentation function and half-cancelled. Either L2 should not be there, or L1's pole needs the same treatment. My reading remains that both cancel against the companion Group II rows.

3. Finite — what L4 becomes once L3 gives back the +1

$$\text{finite} = \xi(1-\xi) \left[\mathcal{X} - d_\perp \mathcal{R} \right] = \xi(1-\xi) \frac{(x-y)^m (x'-y')^{m'}}{(x-y)^2 (x'-y')^2} \left[\delta^{mm'} - d_\perp \hat{r}^m \hat{r}^{m'} \right]$$

Three things are worth noticing about it.

- **It is $\xi(1-\xi)$ — the non-singular part of $\mathcal{P}_{gg}$.** The two singular parts, $\frac{\xi}{[1-\xi]_+}$ (soft) and $\frac{\xi}{\xi}$ (the small- ξ / small- x end), are the ones that went into the BFKL and DGLAP sectors. What is left over is exactly the piece of the splitting function with no endpoint singularity. That is a good consistency check on the split.
- **It is finite by tensor structure, not by accident.** The bracket is traceless:

$$\delta^{mm'} \delta_{mm'} - d_\perp \hat{r}^m \hat{r}^{m'} \delta_{mm'} = d_\perp - d_\perp = 0,$$

while the divergent part of the master integral is pure trace, $\mathcal{G}^{mm'}|_{\text{div}} = \frac{\pi}{2} \log \frac{R^2}{r^2} \delta^{mm'}$. A traceless projector annihilates it. So this block **cannot** carry the logarithm, whatever the kinematics — which is why it is the one sector with no $\frac{1}{\epsilon}$, no μ , no Λ and no ∇ .

- **This is where $d_\perp$ finally earns its keep.** If you had set $d_\perp = 2$ early, the bracket would have been $\delta^{mm'} - 2\hat{r}^m \hat{r}^{m'}$, which is traceless only in exactly two dimensions — the tracelessness would have looked like a coincidence instead of the reason. Keeping $d_\perp$ symbolic is what makes it visible.

4. Two bookkeeping items in the expression as typed

- **Reversed limits.** All of L2, L3, L4 are written $\int_1^{k^+/\nabla}$, and $\frac{k^+}{\nabla} < 1$, so these are reversed integrations: $\int_1^{\xi_0} = -\int_{\xi_0}^1$. The boxed result at the top already has the signs flipped accordingly. Write them all as $\int_{\xi_0}^1$ and carry the signs on the terms — otherwise the non- δ sector silently flips relative to L1.
- **The L2/L3 prefactor.** You now use $\frac{g^2}{4\pi^2}$ (it was $\frac{g^2}{8\pi^2}$ in the earlier version). For L2/L3 to match the $\frac{g^4}{8\pi^4} \frac{N_c}{k^+}$ carried by L1 and L4, the $\frac{g^2}{4\pi^2}$ version requires $\frac{d^3 N_{LO}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+} \int_{x,x',y,y'} \mathcal{X} \mathcal{U}$, the $\frac{g^2}{8\pi^2}$ version requires $\frac{g^2}{\pi^2}$ instead. They differ by a factor 2, so one of the two is wrong — settle it against your LO row.
- Still outstanding from before: delete $\int_{w,w'}$; $\int_{x,y,y'} \rightarrow \int_{x,x',y,y'}$; $U^{ee}(x) \rightarrow U^{ec}(x)$.

5. Summary

sector	your line	what identifies it	fate
BFKL / JIMWLK	L1, unchanged	$\delta(1 - \xi)$; hard scale $k_{\perp}$; one rapidity $\log \log \frac{k^+}{\Lambda}$; LO colour structure	$\log \frac{\sqrt{s}}{\Lambda}$ absorbed into the evolved target; $\log \frac{k^+}{\sqrt{s}}$ stays and its $\sqrt{s}$ -dependence must cancel
DGLAP	L2 + L3, minus the +1	$\frac{d\xi}{\xi^2}$; hard factor at $\frac{k}{\xi}$; $\mathcal{P}_{gg}(\xi)$; parent Wilson lines	open — same pole as L1, so it cannot be treated differently; L3 must be un-factorised first
finite	L4 + the +1 from L3	$\xi(1 - \xi)$, the non-singular part of $\mathcal{P}_{gg}$; traceless tensor $\delta^{mm'} - d_{\perp} \hat{r}^m \hat{r}^{m'}$	none — it is the NLO hard coefficient, finite by construction

Checks: `src/rowG2_sectors3.py` (the three-way split with residual 0, the +1 reassignment, the tracelessness, the L3 obstruction, the normalisation condition), `src/rowG2_advisor.py` (the reversed limits and the single-pole coefficient), `src/rowG2_fate.py` (the BK identity).